

EMERGING TRENDS IN SOFTWARE

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EXECUTIVE SUMMARY



- This capstone project analyses data collected from the 2024 Stack Overflow Developer Survey to identify current technology usage, future technology trends, developer demographics, and employment patterns.
- Web-based languages and technologies continue to be most used and most desired.
- Languages and technologies that are easy to use (simple syntax and typing to avoid errors) are more desired by respondents.

INTRODUCTION



- **Background**

- Organizations require insights into rapidly changing technology trends to make informed decisions regarding hiring, training, and technology investments.
- The Stack Overflow Developer Survey provides valuable information collected from thousands of software professionals worldwide.

- **Project Objectives**

- Collect technology-related data from multiple sources.
- Clean and preprocess the collected data.
- Explore developer demographics and technology preferences.
- Visualize technology trends.
- Build dashboards that communicate insights effectively.

- **Target Audience**

- Business Analysts
- Technology Leaders
- Hiring Managers
- Software Engineers
- Students and Researchers



METHODOLOGY



Data Collection

- The project combines multiple data sources.
- Technology job postings collected using REST APIs
- Popular programming language salary data collected through web scraping
- Stack Overflow Developer Survey dataset

Data Wrangling

- Removed duplicate records
- Identified missing values
- Imputed missing numerical values
- Normalized salary-related fields
- Converted categorical variables where required

Exploratory Data Analysis

- Performed using Python libraries:
- Pandas
- NumPy
- Matplotlib

Dashboard Development

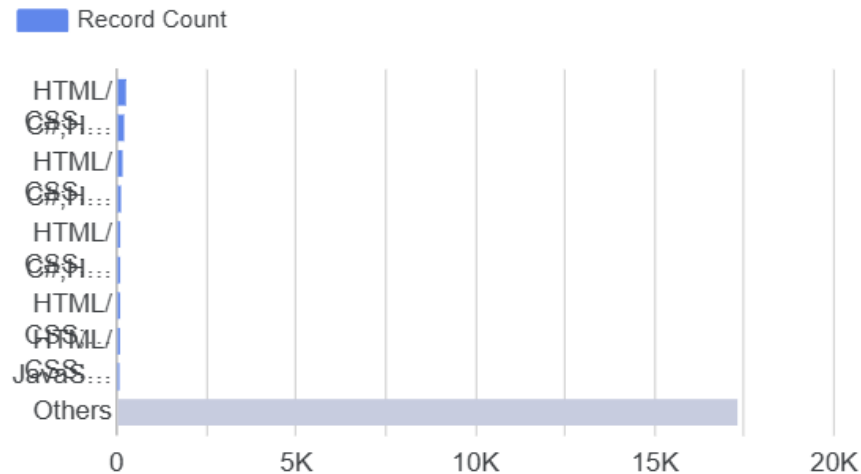
- Interactive dashboards were created using IBM Cognos Analytics to visualize:
- Current Technology Usage
- Future Technology Trends
- Respondent Demographics



PROGRAMMING LANGUAGES TRENDS

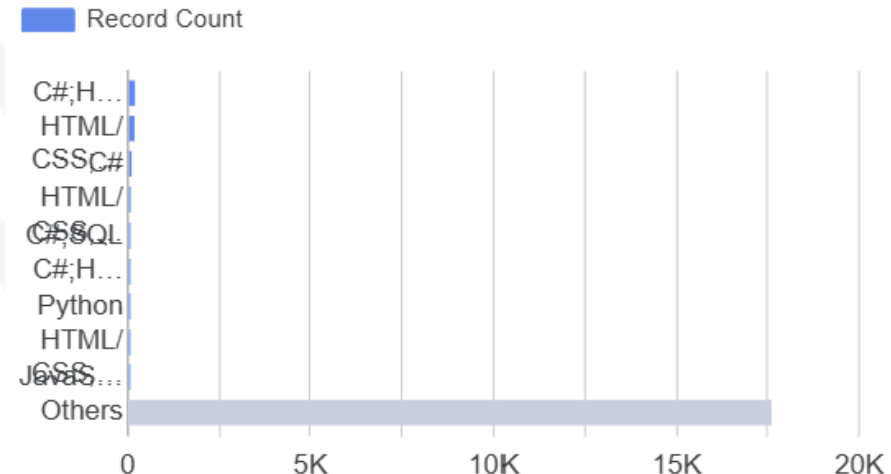
Top 10 Programming Languages – Current Year

Top 10 Languages Used



Top 10 Programming Languages – Next Year

Top 10 Languages Desired Next Year



PROGRAMMING LANGUAGES TRENDS - FINDINGS AND IMPLICATIONS

Findings

- JavaScript continues to dominate professional development.
- Python continues to grow because of AI, automation, and data science.
- TypeScript adoption continues to increase.
- Developers are increasingly interested in modern system programming languages such as Rust.

Implications

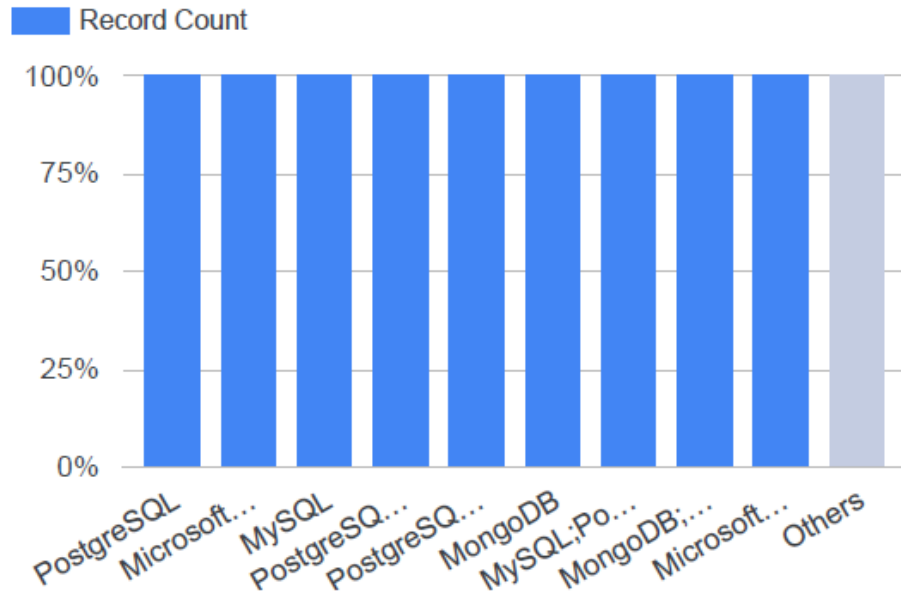
- Organizations should continue investing in JavaScript and Python expertise.
- Learning TypeScript improves employability for web developers.
- Upskilling in modern programming languages helps organizations remain competitive.
- Continuous learning remains essential due to rapidly evolving technology trends.

DATABASE TRENDS

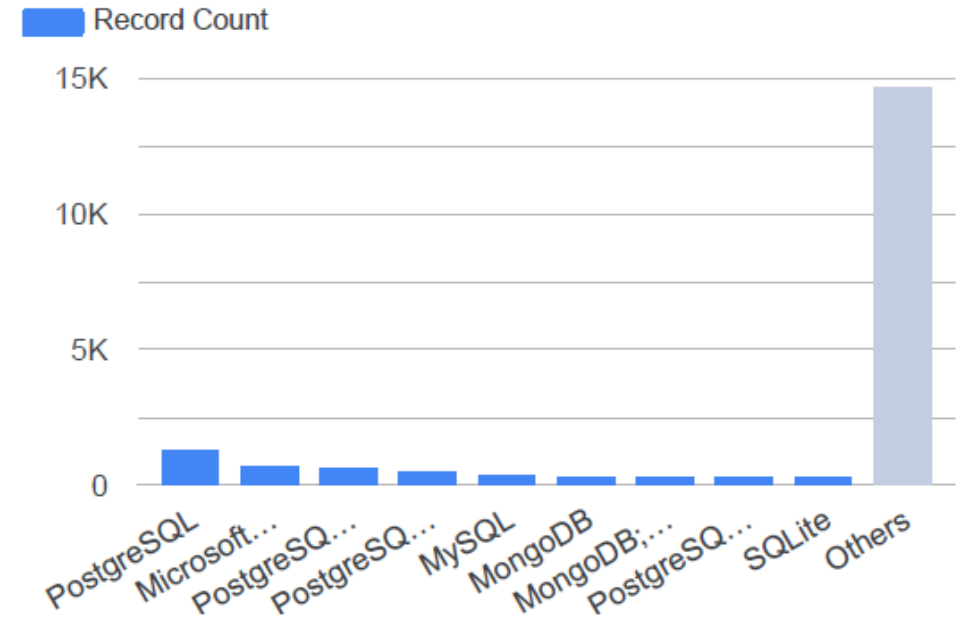
Top 10 Databases Currently in use

Anticipated Future Demand

Top 10 Databases Used



Top 10 Databases Desired Next Year



DATABASE TRENDS - FINDINGS AND IMPLICATIONS

Findings

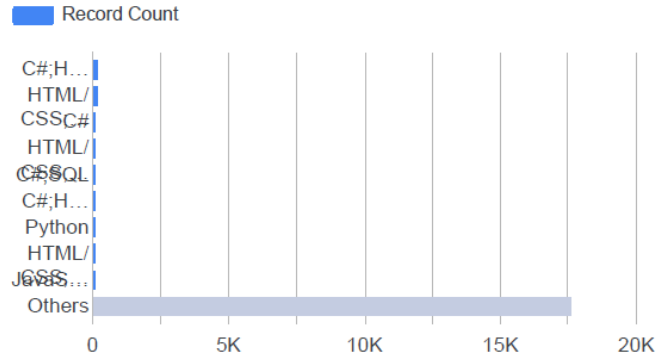
- PostgreSQL remains one of the fastest-growing relational databases.
- MySQL continues to be widely adopted because of its maturity and ecosystem.
- NoSQL databases continue gaining popularity for scalable applications.
- Cloud-native databases are becoming increasingly important.

Implications

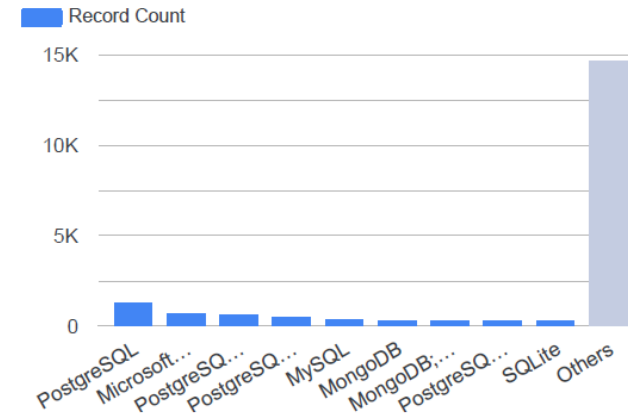
- Organizations should invest in PostgreSQL skills.
- Database administrators need experience with both SQL and NoSQL technologies.
- Cloud database platforms will play a growing role in enterprise applications.
- Developers benefit from familiarity with multiple database technologies.

DASHBOARD TAB 2: Future Technology Trends

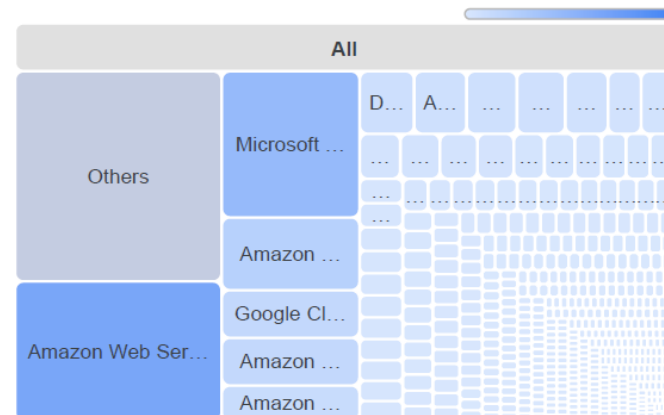
Top 10 Languages Desired Next Year



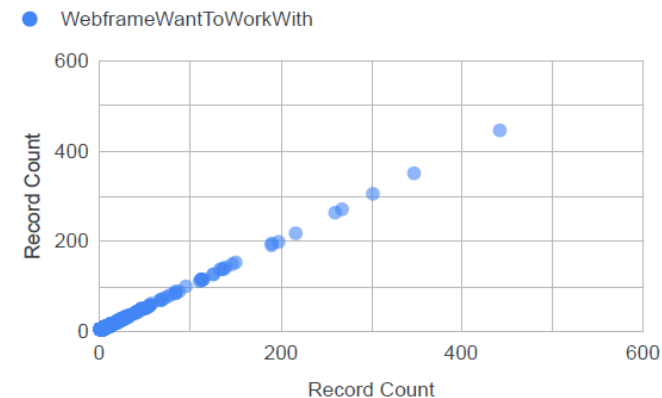
Top 10 Databases Desired Next Year



Top 10 Desired Platforms



Top 10 Desired Web Frameworks



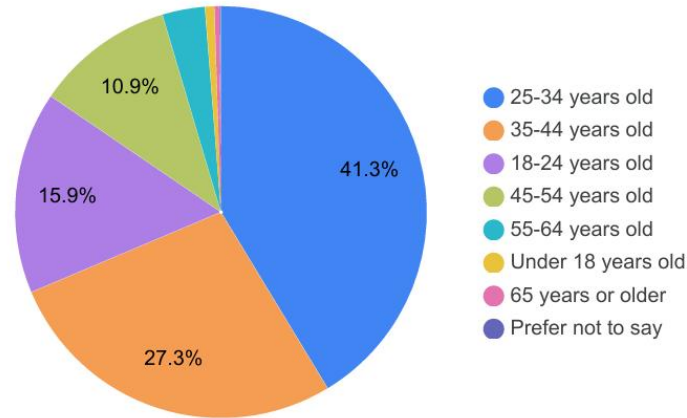
Charts included:

- Top 10 Languages Desired Next Year
- Top 10 Databases Desired Next Year
- Top 10 Desired Platforms
- Top 10 Desired Web Frameworks

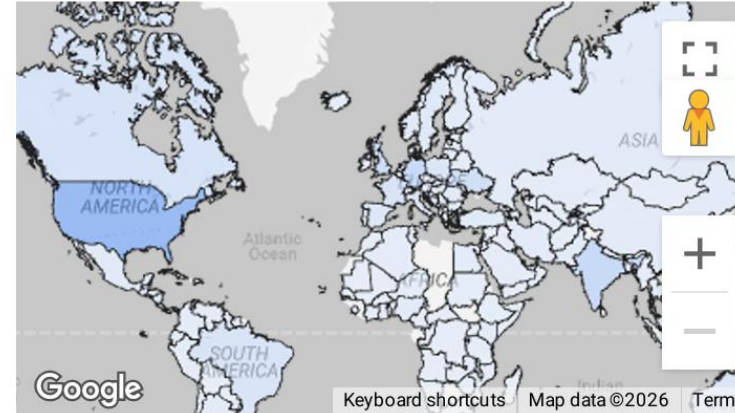


DASHBOARD TAB 3: Demographics

Respondents by Age



Respondent Distribution by Education Level

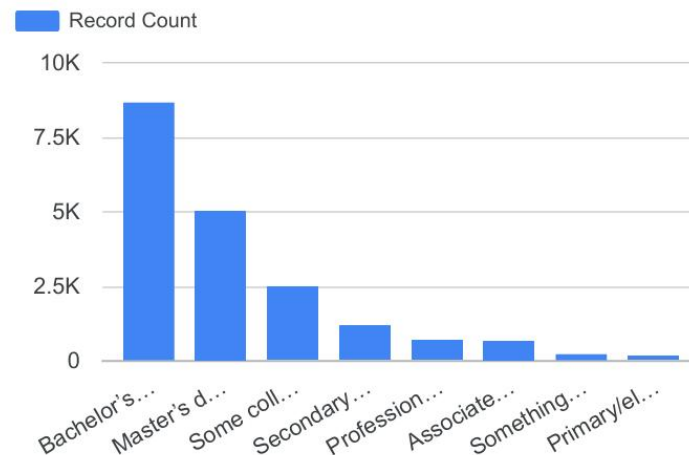


Record Count 1 3,441

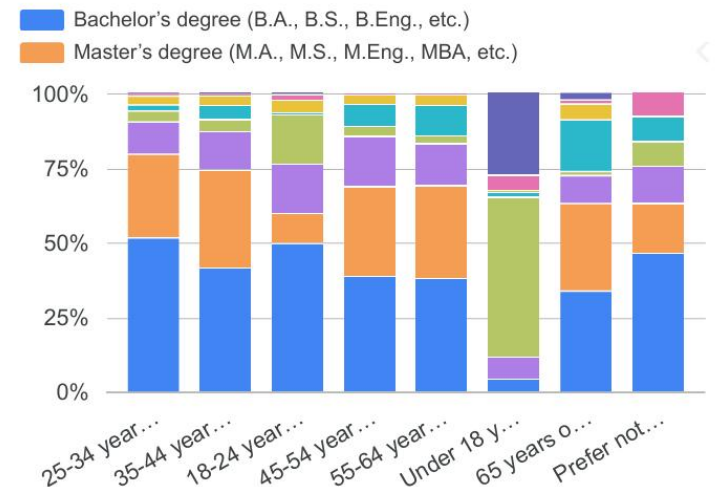
Charts included:

- Respondents by Age
- Respondent Count by Country
- Education Level Distribution
- Age by Education Level

Respondent Distribution by Education Level



Respondent Count by Age Classified by Education Level



DISCUSSION

The analysis demonstrates how data analytics can transform raw survey data into meaningful business insights.

Observations

- Programming languages continue evolving rapidly.
- Cloud computing, artificial intelligence, and automation influence technology choices.
- Traditional technologies remain important despite emerging trends.
- Organizations must continuously adapt to changing developer preferences.

Challenges

- Missing values required preprocessing.
- Multi-select survey responses required additional data transformation.
- Large datasets required efficient visualization techniques.



OVERALL FINDINGS & IMPLICATIONS

Major Findings

- JavaScript and Python remain dominant programming languages.
- PostgreSQL is becoming increasingly popular.
- Developers show growing interest in cloud-native technologies.
- Younger professionals continue driving technology adoption.
- AI-related technologies are becoming mainstream.

Business Implications

- Organizations should continue investing in cloud technologies.
- Continuous employee upskilling is essential.
- Hiring strategies should align with future technology trends.
- Data-driven decision-making improves workforce planning.
- Interactive dashboards provide valuable decision support.

CONCLUSION



Conclusion

- This capstone project demonstrated the complete data analytics lifecycle—from data collection and cleaning to analysis, visualization, and dashboard development.

Project Achievements

- Successfully collected data using APIs and web scraping.
- Cleaned and prepared the dataset through data wrangling.
- Performed exploratory data analysis using Python.
- Created informative visualizations to identify trends.
- Built an interactive IBM Cognos Analytics dashboard.

Final Takeaway

- Data visualization and dashboarding enable organizations to transform complex datasets into actionable insights that support informed business decisions.

